

Exp. No 08

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ESP8266 - LED Control via Mobile Interface

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- **Components Required**

Sr.No.	Component	Quantity	Description	Use
1.	ESP8266	1	It is a Wi-Fi enabled microcontroller board.	It is used to connect to the mobile and control the LED through a web page
2.	Mobile	1	It is a device used to access the web interface.	It is used to control the LED wirelessly.

- **Circuit Diagram**



- **Program Code**

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>

const char* apName = "ESP8266_Control";
const char* apPassword = "12345678";

ESP8266WebServer server(80);

// Built-in LED on most NodeMCU ESP8266 boards
#define LED_PIN 2

String webpage() {
  String html = "<!DOCTYPE html>";
  html += "<html>";
  html += "<head>";
  html += "<meta name='viewport' content='width=device-width, initial-scale=1'>";
  html += "<title>ESP8266 Control</title>";

  html += "<style>";
  html += "body {";
  html += "font-family: Arial;";
  html += "text-align: center;";
  html += "margin-top: 60px;";
  html += "}";
  html += "button {";
  html += "font-size: 25px;";
  html += "padding: 15px 35px;";
  html += "margin: 10px;";
  html += "}";
  html += "</style>";

  html += "</head>";
  html += "<body>";

  html += "<h1>ESP8266 LED Control</h1>";

  html += "<p>";
  html += "<a href='/on'><button>LED ON</button></a>";
  html += "</p>";

  html += "<p>";
  html += "<a href='/off'><button>LED OFF</button></a>";
  html += "</p>";
```

```

    html += "</body>";
    html += "</html>";

    return html;
}

void setup() {

    Serial.begin(115200);

    pinMode(LED_PIN, OUTPUT);

    // LED OFF
    digitalWrite(LED_PIN, HIGH);

    // Create ESP8266 hotspot
    WiFi.softAP(apName, apPassword);

    Serial.println();
    Serial.println("ESP8266 Hotspot Started!");

    Serial.print("Wi-Fi Name: ");
    Serial.println(apName);

    Serial.print("Password: ");
    Serial.println(apPassword);

    Serial.print("IP Address: ");
    Serial.println(WiFi.softAPIP());

    // Main page
    server.on("/", []() {
        server.send(200, "text/html", webpage());
    });

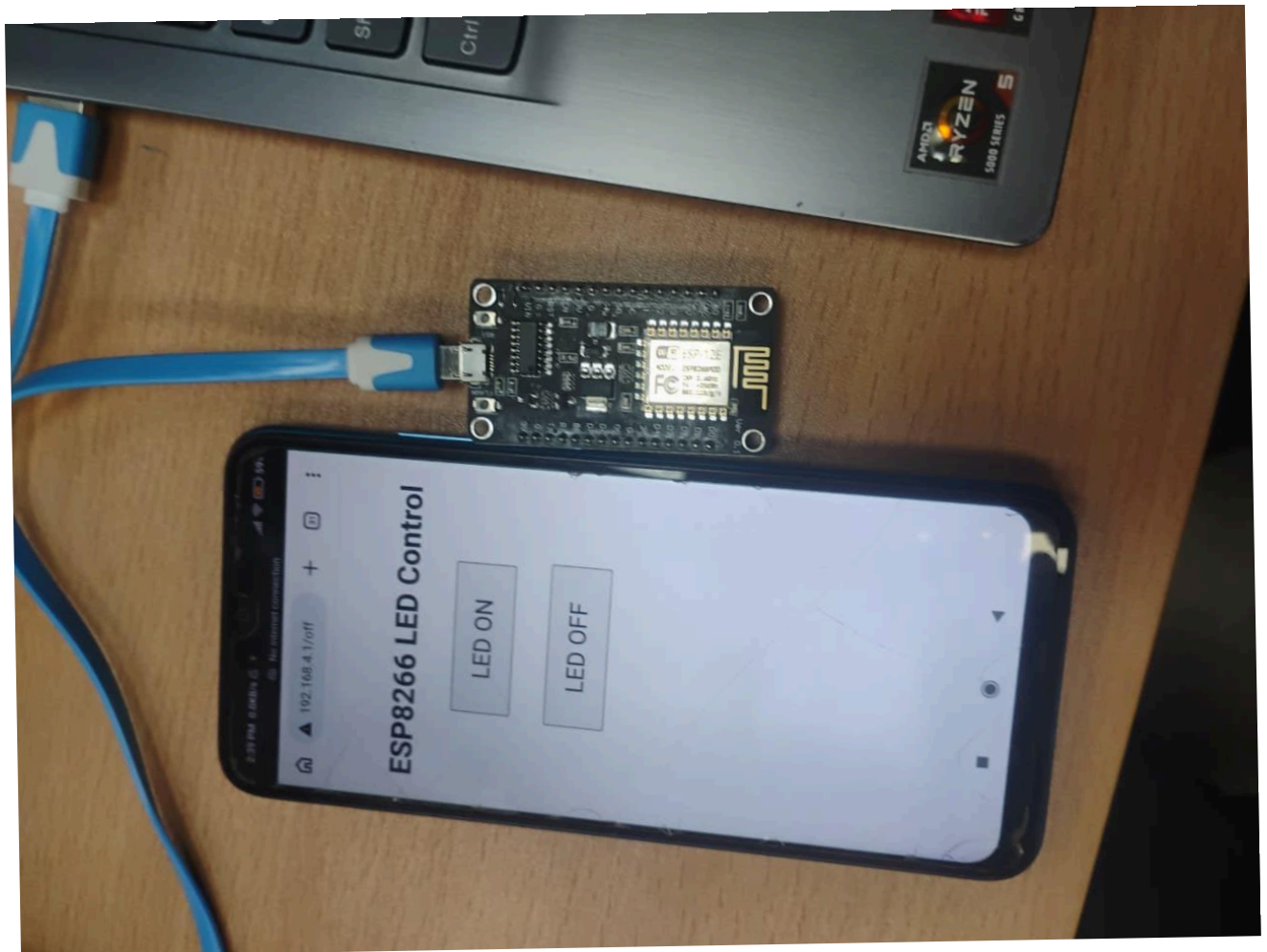
    // LED ON
    server.on("/on", []() {
        digitalWrite(LED_PIN, LOW);
        server.send(200, "text/html", webpage());
    });

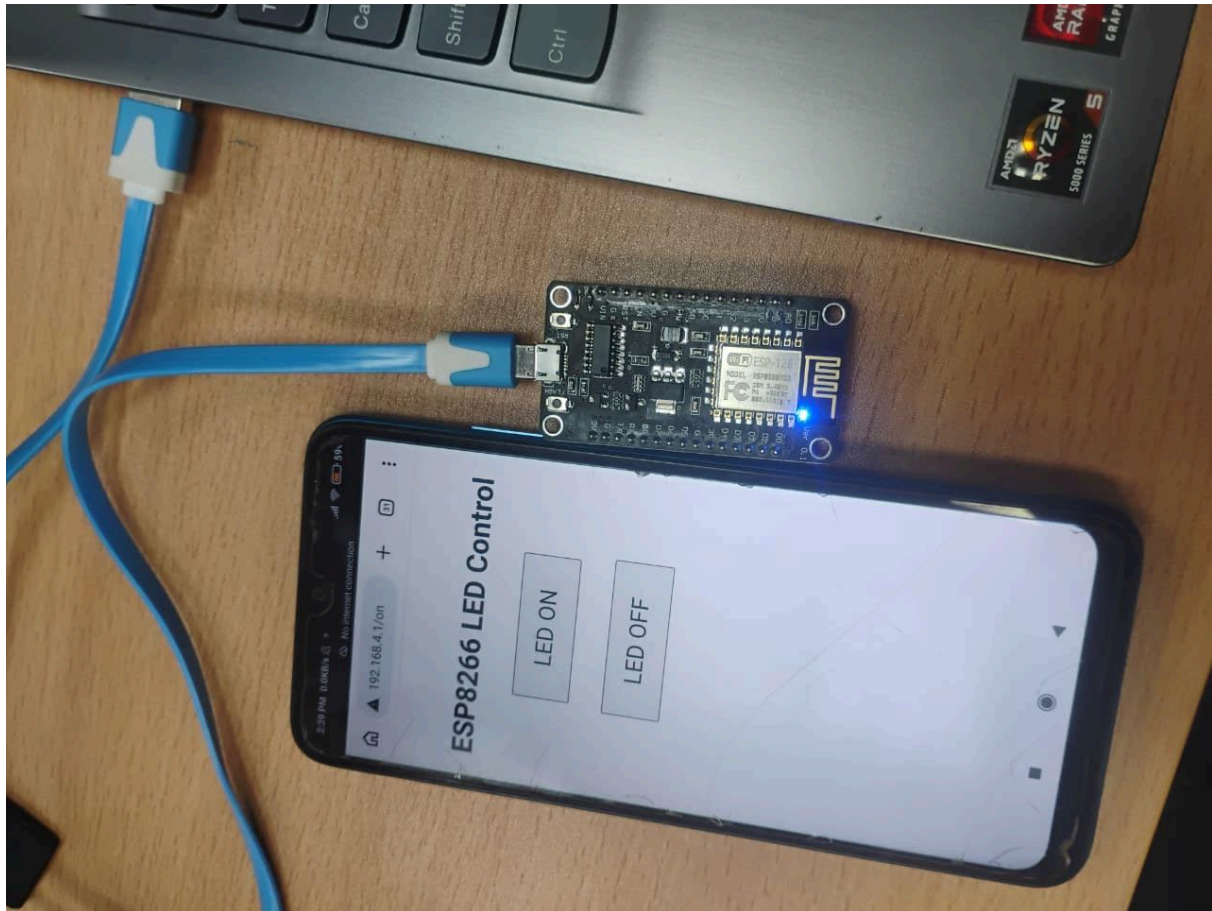
    // LED OFF
    server.on("/off", []() {

```

```
digitalWrite(LED_PIN, HIGH);  
server.send(200, "text/html", webpage());  
});  
  
server.begin();  
  
Serial.println("Web server started!");  
}  
  
void loop() {  
  server.handleClient();  
}
```

- **Photograph of the Experiment**





- **Conclusion**

We learned how to control the built-in LED of ESP8266 using a mobile phone. We created a simple web interface on the mobile to turn the LED ON and OFF. This experiment helped us understand how Wi-Fi can be used to control devices wirelessly.